

AUTHENTICLENS

KNOWLEDGE BASE - DOCUMENT 6 OF 9

Document 6 - Nano Banana Execution Playbook

Prompt Architecture, Editing Commands, Negative Constraints, Repair Loops, and
Quality Control

Purpose: upload-ready operational documentation for the AuthenticLens Gemini Gem. This document translates theory into retrieval-friendly procedures, decision rules, prompt modules, and quality-control checks.

Prepared for: AuthenticLens - The Realism Architect

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Language: English (Canada)

Primary target: Gemini Pro + Nano Banana / Nano Banana Pro image generation and editing workflows

Purpose

Convert AuthenticLens analysis into precise Gemini image-generation and image-editing instructions. This document is optimized for instruction-following models using text+image context, especially Nano Banana and Nano Banana Pro. It treats “Nano Banana 2 Pro” as user shorthand, not an official model name.

Intended Gemini Behaviour

- Use natural-language edit commands rather than parameter-heavy legacy syntax.
- State preservation constraints before creative changes.
- Ask the image model for targeted edits, not full regeneration, when preserving identity matters.
- Use negative constraints sparingly and specifically.
- After output, run the QC checklist and repair only what failed.

A. Master Prompt Architecture

Prompt layer	What to write	Example
1. Task command	State the exact operation.	Enhance the attached photo for realism without changing the subject.
2. Preservation constraints	Lock face, body, complexion, clothing, and background unless requested.	Preserve the subject’s facial identity, age, skin tone, body structure, hairstyle, expression, and pose.
3. Subject description	Describe visible subject neutrally.	Adult man standing outdoors, frontal 3/4 pose, dark jacket.
4. Scene/background	Specify keep/change and context.	Keep current background, or replace with Rideau Street, Ottawa, early spring.
5. Lighting	Define direction, quality, and integration.	Natural overcast daylight, soft shadows, realistic contact shadow under shoes.
6. Camera/lens behaviour	Use plausible camera style.	Real smartphone photo, mild lens distortion, no fake cinematic bokeh.
7. Texture/material realism	Specify skin, fabric, hair, metal, pavement.	Restore skin pores, fabric weave, stitching, natural hair edges.
8. Editing style	Define restrained finishing.	Subtle Lightroom-style colour correction, natural contrast, no over-HDR.
9. Negative constraints	List only relevant failure preventions.	Do not alter the face, do not change complexion, no plastic skin, no warped hands.
10. Final QC instruction	Force model to prioritize believable output.	Before finalizing, verify identity, hands, shadows, perspective, and material realism.

B. Preservation-First Editing Patterns

User intent	Prompt pattern
Enhance this image without changing the subject	Improve realism, clarity, lighting balance, and photographic texture only. Preserve the subject exactly: same face, skin tone, body, hair, expression, clothing, pose, and background composition.
Do not alter the face	The face is locked. Do not modify facial structure, eye shape, nose, mouth, jawline, cheeks, skin tone, age, expression, or facial hair. Only improve lighting/texture around it.
Keep complexion accurate	Maintain the subject's natural skin complexion. Correct only exposure or white balance. Do not lighten, darken, recolour, or beautify skin unless explicitly requested.
Only improve realism	Do not add new objects, clothing, background elements, logos, or dramatic effects. Apply subtle real-camera imperfections and correct physics.
Do not change clothing/background unless requested	Keep wardrobe and environment unchanged. Do not replace garments, logos, setting, storefronts, vehicles, or sky.

C. Realism Prompt Modules

Module	Instruction
Realistic portrait enhancement	Preserve the person exactly. Improve natural skin texture with visible pores and subtle under-eye detail, reduce plastic smoothing, correct white balance, refine hair edges, and add realistic soft shadow transitions. Avoid beauty retouching or identity drift.
Realistic street photo	Make the scene read as a real candid street photograph: coherent horizon, plausible pedestrians/vehicles, natural daylight, slight lens imperfection, real pavement texture, readable signage only where clearly rendered, and no impossible architecture.
Fashion editorial	Create restrained high-end fashion realism with believable garment drape, seams, fabric weight, subtle studio or street lighting, natural skin texture, and no fake glossy render finish.
Luxury outfit swap	Replace only the clothing with a minimal Saint Laurent-style luxury outfit: clean tailoring, premium fabric, understated black/neutral palette, realistic seams and folds. Keep the same face, body, skin tone, pose, hands, and background light.
E-commerce product photo	Enhance product clarity for web: true colour, sharp fabric weave, visible stitching, clean

	zipper/hardware, realistic studio shadow, no over-sharpened halos, no invented product details.
Background replacement	Replace the background while matching original camera height, lens perspective, depth of field, light direction, colour temperature, contact shadows, and grain. Subject must be physically grounded.
Group collage	Composite the subjects together while preserving each face and appearance. Match scale, head height, lens compression, colour grade, grain, shadow direction, and edge softness.
AI-artifact correction	Target only visible artifacts: fix hands, text, melted edges, fabric seams, shadows, reflections, or perspective. Do not regenerate the whole person or scene.
Lightroom-style realism	Apply restrained real-photo edits: balanced exposure, recovered highlights, natural shadows, moderate contrast, subtle texture, no heavy HDR, no waxy smoothing, no excessive saturation.
Phone-camera realism	Use modern smartphone capture characteristics: realistic HDR tone mapping, slight edge enhancement, mild denoise in shadows, deep focus, subtle lens distortion, no impossible DSLR shallow depth unless requested.
35mm film realism	Use 35mm film cues subtly: fine grain, slight highlight rolloff, modest lens falloff, minor colour response, natural imperfections. Do not make the image muddy or overly vintage.
Ottawa spring street realism	Use early spring Ottawa street cues: cool daylight, damp or dry pavement consistent with weather, bare/budding trees, layered but not winter-heavy clothing, accurate road/sidewalk scale, no invented impossible landmarks.

D. Nano Banana Negative Instruction Library

- No AI glow, plastic skin, waxy face, doll-like eyes, over-smooth gradients, or hyper-glossy render lighting.
- No altered identity, no changed face, no changed complexion, no age shift, no body reshaping, no beautification unless requested.
- No warped hands, fused fingers, distorted limbs, broken wrists, melted ears, fake teeth, or impossible anatomy.
- No melted fabric, painted-on seams, floating zipper, fake stitching, or weightless clothing.
- No unreadable text, fake logos, pseudo-brand marks, corrupted signage, or invented store names unless blurred/background-only.
- No inconsistent shadows, floating subject, missing contact shadow, mismatched lighting, or impossible reflections.

- No over-sharpening, oversmoothing, overdone HDR, fake bokeh, noisy overlay, excessive chromatic aberration, or vignette that looks like a filter.
- No impossible perspective, duplicated objects, repeating textures, synthetic-looking background, or cutout halos.

E. Quality-Control Checklist

- Face unchanged?
- Skin tone unchanged unless requested?
- Hands and limbs anatomically correct?
- Clothing physically believable?
- Fabric, zipper, stitching, and logos clean?
- Subject grounded with contact shadows?
- Background matches lens, scale, light, season, and geography?
- Text/logos readable or safely avoided?
- No overprocessing?
- Result looks like a plausible photograph?

F. Repair Loop

1. Identify the specific failure: identity, anatomy, lighting, perspective, fabric, text, compression, background, or overprocessing.
2. Classify the failure type: AI artifact, editing artifact, composite mismatch, prompt ambiguity, or model overreach.
3. Rewrite the prompt with stricter preservation constraints and fewer creative instructions.
4. Apply targeted correction only to the failed region or property.
5. Re-evaluate against the QC checklist.
6. Stop once the issue is corrected. Do not keep adding realism effects.

G. Ready-to-Use Output Templates

Request	Prompt
Enhance this picture	Enhance the attached image for realistic photographic quality. Preserve the subject exactly: same face, body, skin tone, hairstyle, expression, pose, clothing, and background. Improve exposure, natural contrast, texture, sharpness, and subtle real-camera imperfections. Do not beautify, reshape, or regenerate the subject.
Make more realistic / less AI-generated	Reduce synthetic appearance by correcting lighting, shadows, material texture, skin detail, edge realism, and lens consistency. Add only subtle camera-realistic grain/noise if needed. Avoid plastic skin, AI glow, fake bokeh, oversaturation, warped hands, and melted fabric.
Rideau Street Ottawa spring background	Replace the background with a realistic Rideau

	Street, Ottawa, Canada street scene in early spring. Match camera height, lens perspective, daylight, shadows, pavement, road/sidewalk scale, and seasonal cues. Keep the subject identical and grounded with realistic contact shadows.
Saint Laurent clothing swap	Change only the clothing to a realistic Saint Laurent-style luxury outfit: minimal tailoring, premium dark fabric, clean understated design. Preserve the same face, complexion, body, pose, hands, and lighting. Avoid fake unreadable logos or counterfeit-looking marks.
Melato product photo sharpening	Enhance this product photo for melato.ca. Preserve the product design and colour exactly. Increase premium clarity, fabric weave visibility, zipper/hardware detail, stitching definition, clean shadow, and true colour. Avoid over-sharpening halos or invented fabric patterns.
Natural group collage	Create a natural blended group image using the provided subjects. Do not alter any face or appearance. Match scale, camera angle, lighting, shadows, grain, edge softness, and background integration so they look photographed together.
Analyze real/AI/edited/composite	Classify the image using four categories: likely real, real but heavily edited, likely AI-generated, or composite/manipulated. Separate structural AI artifacts from Lightroom/editorial artifacts and report confidence plus uncertainty.
Fix perspective/alignment	Correct perspective, horizon, vertical alignment, lens distortion, and image tilt while preserving all subjects and objects. Do not crop aggressively or alter identity.
Complexion adjustment by reference	Adjust skin complexion subtly to match the provided reference image. Preserve facial identity and natural skin texture. Treat the change as exposure/white-balance correction, not a new skin tone.

Quick Reference Summary

Best execution pattern: task + preservation + target edit + camera/light + texture/material + restrained finish + negative constraints + QC. If the output fails, repair the specific failure only.

Source Notes and Reliability Labels

Use source notes as operational grounding, not as a claim of legal or forensic certainty. If metadata, original files, or verified provenance are unavailable, AuthenticLens must label final authenticity claims as probabilistic.

- **Internal Doc 1:** AuthenticLens_Doc1_AI_Artifact_Lexicon.pdf - primary artifact taxonomy and detection guidance.
- **Internal Doc 2:** AuthenticLens_Doc2_Photographic_Physics.pdf - optical physics, sensor/film artifacts, lighting, exposure, and colour foundations.
- **Internal Doc 3:** AuthenticLens_Doc3_Realism_Prompting_Formulas.pdf - realism prompt formulas, camera/film language, and platform prompting references.
- **Google AI Developers:** Image generation with Gemini / Nano Banana and Nano Banana Pro documentation. Source note: confirms Gemini image generation/editing is conversational, supports text+image inputs, identifies Nano Banana as gemini-2.5-flash-image and Nano Banana Pro as gemini-3-pro-image-preview, and states generated images include SynthID watermarking.
- **Adobe Lightroom Help:** Lightroom documentation for Texture, Clarity, Dehaze, Vignette, Grain, Sharpening, Noise Reduction, local adjustments, and masking. Source note: used to separate editing artifacts from AI-generation artifacts.
- **C2PA Specification:** C2PA Content Credentials specification and explainer. Source note: provenance is represented by signed manifests / Content Credentials that can record creation and modification assertions.
- **NIST Digital Media / Deepfake Guidance:** NIST digital and multimedia evidence resources and deepfake guidance. Source note: provenance, supply chain, compression, and context matter; visual inspection alone is not definitive.
- **W3C sRGB:** sRGB as the default colour space for web images. Source note: used for e-commerce and social-media colour delivery guidance.

Reliability labels used in this knowledge base: Confirmed source, Common photography practice, Image-forensics heuristic, Operational guidance, Prompting heuristic, Model-behaviour observation.